

L12a: How Do Micro Decisions Lead to Emergent Macro Trends?

CHEME 5660 Cornell University

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today, we write down simple rules for individual traders, simulate many of them together, and measure what the market as a whole does.

Today's concepts

- Agent-based models: the five parts every model shares, in the Farmer–Joshi market and the minority game.
- The Boltzmann choice rule: the maximum-entropy choice, with a temperature that sets how strongly agents favor high scores.
- Micro to macro: fat tails and volatility clustering in the Farmer–Joshi market; volatility in the minority game.

What You Need to Know

Core concepts

- Five parts: agents, information, a decision rule, an aggregate, and feedback from the aggregate to the next decisions.
- Maximum entropy for a fixed expected score gives $p_j = \exp(U_j/T)/\mathcal{Z}$: greedy as $T \to 0$, uniform as $T \to \infty$.
- Farmer–Joshi value investors and trend followers together produce fat tails and volatility clustering; the minority game's volatility per agent depends on α and T.

For understanding (you won't reproduce these)

- The Farmer–Joshi threshold rules and our parameter values.
- The minority-game price, an extension of the original model.

Lectures and Examples

Lecture:
CHEME-5660-L12a-Lecture-AgentBasedMarkets-Fall-2026.ipynb

Examples:
CHEME-5660-L12a-Example-FarmerJoshi-Fall-2026.ipynb
CHEME-5660-L12a-Example-ThermalMinorityGame-Fall-2026.ipynb

In the News: The November 5 Sell-Off

On Thursday, November 5, 2026, the Nasdaq fell 5.5% and the S&P 500 about 4.4%, the steepest one-day declines since April 2025, on worries that AI data-center spending could slow:

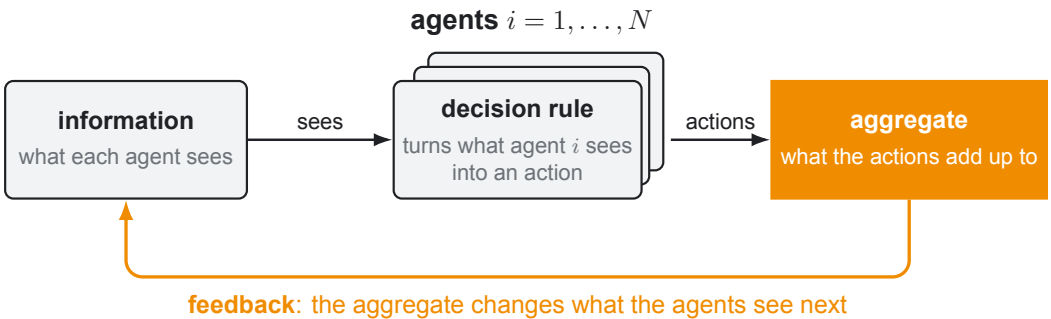
- The seven largest technology companies: lost more than 1 trillion USD of market value.
- Retail investors: sold a net 338 million USD of individual stocks, the largest one-day outflow since March 2020. “On a typical sell-off, retail buy the dip. Today, that bid has effectively disappeared from individual names.”

Connection: the day’s move was the sum of individual buy, sell, or wait decisions. On 2008: “the fault lies not in the stars, but in us.”

Source: CNBC; CNBC, Vanda Research data; FCIC report

Five Parts of Every Agent-Based Model

Agents see information, their decision rules turn it into actions, the actions add up to an aggregate, and the aggregate feeds back into what they see next:



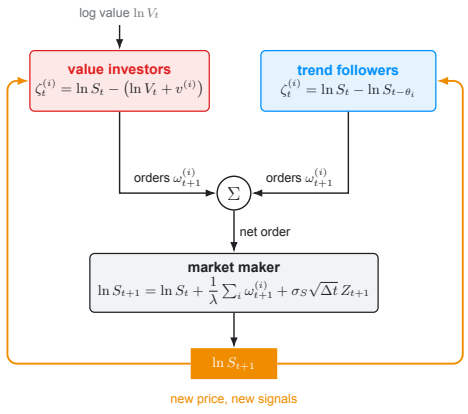
The Structure of an Agent-Based Model

An agent-based model describes individual decision makers and their rules, simulates them together, and measures the market-wide result:

Part	Farmer–Joshi market	Minority game
Agents	Value investors, trend followers	N agents, J fixed policies each
Information	Mispricing or price change over a lag	History μ_t of the last m outcomes
Decision rule	Enter or exit at a threshold	Choose a policy by its score
Aggregate	Net order, then the price	Attendance A_t
Feedback	New price changes every signal	A_t updates scores and history

The Farmer–Joshi Market: One Step

Signals set positions, positions set orders, and the net order sets the price, which changes every signal at the next step:



Value Investors and Trend Followers

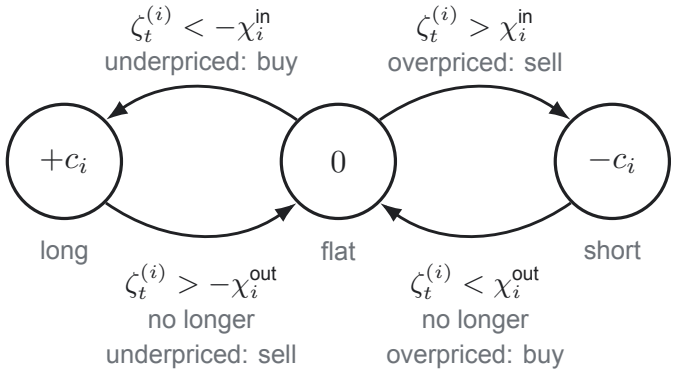
Each trader enters at a signal beyond $\pm\chi_i^{\text{in}}$ and exits back across $\pm\chi_i^{\text{out}}$. The position size is $c_i = a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$, where $a > 0$ is a constant scale factor:

- Value investors: signal $\ln S_t - (\ln V_t + v^{(i)})$, where $\ln V_t$ is the log of the fundamental value, a random walk, and $v^{(i)}$ is the investor's fixed offset; sell when overpriced, buy when underpriced.
- Trend followers: signal $\ln S_t - \ln S_{t-\theta_i}$; buy after a rise, sell after a fall.

Source: Farmer and Joshi (2002)

The Threshold Rule

A value investor's position is one of three states. It enters when the mispricing passes $\pm\chi_i^{\text{in}}$ and exits when it comes back past $\pm\chi_i^{\text{out}}$:



The Market Maker Sets the Price

Trader i holds $x_t^{(i)} \in \{-c_i, 0, +c_i\}$ and orders $\omega_{t+1}^{(i)} = x_{t+1}^{(i)} - x_t^{(i)}$. The market maker moves the log price in proportion to the net order:

$$\ln S_{t+1} = \ln S_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \sigma_S \sqrt{\Delta t}\, Z_{t+1}$$

where λ is the liquidity, $Z_{t+1} \sim \mathcal{N}(0,1)$ is independent noise, σ_S is its annualized volatility, and each step lasts Δt years.

Growth Rate: Farmer–Joshi and GBM

The growth rate over step t, $g_{t+1} = \frac{1}{\Delta t} \ln \frac{S_{t+1}}{S_t}$, follows directly from the price rule. Let's compare it with single-asset geometric Brownian motion:

$$\text{Farmer–Joshi:} \quad g_{t+1} = \underbrace{\frac{1}{\lambda\,\Delta t} \sum_i \omega_{t+1}^{(i)}}_{\text{traders}} + \underbrace{\frac{\sigma_S}{\sqrt{\Delta t}} Z_{t+1}}_{\text{price noise}}$$

$$\text{Geometric Brownian motion:} \quad g_{t+1} = \underbrace{\mu_g}_{\text{constant}} + \underbrace{\frac{\sigma}{\sqrt{\Delta t}} Z_{t+1}}_{\text{price noise}}$$

where $\sigma = \sigma_g \sqrt{\Delta t}$ is the annualized volatility and $\mu = \mu_g + \frac{1}{2}\sigma^2$ is the drift. We estimate the mean μ_g and the standard deviation σ_g of the growth rate from data.

In geometric Brownian motion, the first term is a constant that we estimate from data. In the Farmer–Joshi market, the first term comes from a mechanistic model of how the actions of the traders change the growth rate.

The Farmer–Joshi Algorithm

Initialize: thresholds $\chi_i^{\text{in}} > \chi_i^{\text{out}}$, offsets $v^{(i)}$, lags θ_i, a, λ, σ_S, σ_v, Δt, steps R;
$c_i \leftarrow a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$; $x_0^{(i)} \leftarrow 0$. For each step $t = 0, \dots, R-1$ **do**:

1. Signals: $\zeta_t^{(i)} \leftarrow \ln S_t - (\ln V_t + v^{(i)})$ (value) or $\ln S_t - \ln S_{t-\theta_i}$ (trend, once $t \geq \theta_i$).
2. Positions: the threshold rule sets $x_{t+1}^{(i)} \in \{-c_i, 0, +c_i\}$.
3. Orders: $\omega_{t+1}^{(i)} \leftarrow x_{t+1}^{(i)} - x_t^{(i)}$.
4. Price: draw $Z_{t+1} \sim \mathcal{N}(0,1)$; $\ln S_{t+1} \leftarrow \ln S_t + \frac{1}{\lambda}\sum_i \omega_{t+1}^{(i)} + \sigma_S\sqrt{\Delta t}\, Z_{t+1}$.
5. Value: draw $Z'_{t+1} \sim \mathcal{N}(0,1)$; $\ln V_{t+1} \leftarrow \ln V_t + \sigma_v\sqrt{\Delta t}\, Z'_{t+1}$.

Output: log prices, log values, growth rates $g_{t+1} = \frac{1}{\Delta t}\ln\frac{S_{t+1}}{S_t}$, volume.

Stylized Facts: Simulated and Real

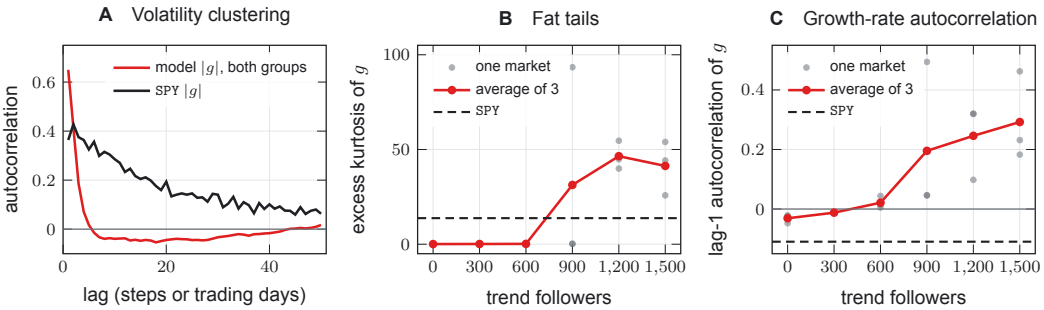
Both groups together, against `SPY` daily growth rates (kurtosis and autocorrelations do not depend on units):

Statistic	Simulated	`SPY`
Excess kurtosis	38.8	13.8
Autocorrelation of g, lag 1	0.31	-0.11
Autocorrelation of $\|g\|$, lag 1	0.65	0.36
Autocorrelation of $\|g\|$, lag 5	0.02	0.33
Autocorrelation of volume, lag 1	0.55	0.75

Fat tails and clustering come out of the rules, but the clustering fades within a few steps.

Clustering and the Mix of Traders

The model's clustering fades within a few steps (A). Fat tails and a positive growth-rate autocorrelation appear only with enough trend followers (B, C); every mix misses SPY by a wide margin on at least one statistic:



The Minority Game

Each round, N (odd) agents buy ($+1$) or sell (-1); the minority side, $-\text{sign}\, A_t$, wins. Agent i takes the action $a_{i,t} = \pi_{ij}(\mu_t)$ of one of its J policies, tables $\pi_{ij}(\mu)$ over the $P = 2^m$ histories μ:

$$A_t = \sum_{i=1}^{N} a_{i,t} \qquad\qquad U_{ij,t+1} = U_{ij,t} - \frac{\pi_{ij}(\mu_t) A_t}{P}$$

where the attendance A_t is the number of buyers minus sellers and $U_{ij,t}$ is the score of policy π_{ij}. A policy that would have put its agent in the minority gains score.

Source: Challet and Zhang (1997); scores divided by P as in Challet, Marsili, and Zecchina (2000)

Volatility per Agent

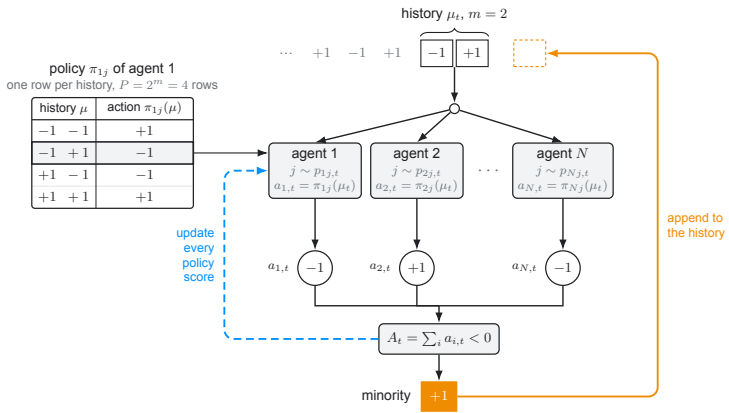
Each coin flip is a Bernoulli trial: $X_{i,t} = 1$ (buy) with probability $1/2$, and its action $a_{i,t} = 2X_{i,t} - 1$ has mean 0 and variance 1. The attendance adds the N agents' independent actions, so it has variance N, the baseline for the volatility per agent:

$$\frac{\sigma_A^2}{N} = \frac{\langle A^2 \rangle}{N} \qquad\qquad \alpha = \frac{P}{N}$$

where $\langle \cdot \rangle$ averages over rounds and α is the number of histories per agent. $\sigma_A^2/N > 1$: more agents on the same side than with coin flips; $\sigma_A^2/N < 1$: a more even split. In the greedy game, each agent uses its best-scoring policy. With $J = 2$ and α below about 0.2, the game is crowded: many agents switch to the same action at the same time.

One Round of the Minority Game

The history goes to every agent; the actions add up to the attendance, which feeds back into the history and every policy's score:



Adding a Price: Instructor's Extension

The original game has no price. A market maker treats the attendance as the excess demand and moves the log price, as in the Farmer–Joshi market:

$$\ln S_{t+1} = \ln S_t + \frac{A_t}{\lambda} + \sigma_S \sqrt{\Delta t}\, Z_{t+1}$$

where λ is the liquidity, $Z_{t+1} \sim \mathcal{N}(0,1)$ is independent noise, and σ_S is its annualized volatility. If one round lasts Δt years, the growth rate is $g_{t+1} = \frac{1}{\Delta t} \ln \frac{S_{t+1}}{S_t}$, with mean square about $\sigma_A^2/(\lambda^2 \Delta t^2) + \sigma_S^2/\Delta t$. At fixed N, λ, and σ_S, a crowded game moves the price more.

Related: Challet, Chessa, Marsili, and Zhang (2000), with the opposite sign and no noise

The Boltzmann Choice Rule

An agent with scores $U_1, \dots, U_J$ uses policy π_j with probability:

$$\boxed{p_j = \frac{1}{\mathcal{Z}} \exp\left(\frac{U_j}{T}\right)}, \qquad \mathcal{Z} = \sum_{k=1}^{J} \exp\left(\frac{U_k}{T}\right)$$

where $T > 0$ is the temperature, in the units of the scores.

Among all choice probabilities with a given expected score, this rule has the largest entropy, a measure of how spread out the probabilities are.

Source: Cavagna, Garrahan, Giardina, and Sherrington (1999); proof in the optional advanced notebook

What the Temperature Does

The temperature sets how strongly the agent favors high scores:

- $T \to 0$: the best policy has probability 1, the greedy minority game.
- $T \to \infty$: every policy has probability $1/J$.
- In between: the best policy has the largest probability, and the other policies share the rest in proportion to $\exp(U_j/T)$.

The Thermal Minority Game Algorithm

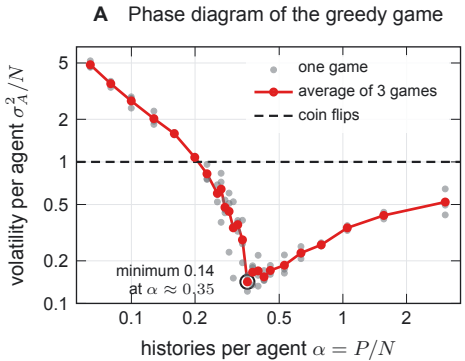
Initialize: N (odd), m, J, T, rounds R, λ, σ_S, Δt; $P = 2^m$; random tables $\pi_{ij}(\mu)$; scores $U_{ij,0} \leftarrow 0$; random μ_0; starting price S_0. For each round $t = 0, \dots, R-1$ **do**:

1. Choose: $p_{ij,t} = \exp(U_{ij,t}/T)/\mathcal{Z}_{i,t}$; draw $j_{i,t} \sim \text{Categorical}(p_{i1,t}, \dots, p_{iJ,t})$.
2. Act: $a_{i,t} \leftarrow \pi_{i\, j_{i,t}}(\mu_t)$.
3. Aggregate: $A_t \leftarrow \sum_i a_{i,t}$; the minority action is $-\text{sign}\, A_t$.
4. Score: $U_{ij,t+1} \leftarrow U_{ij,t} - \pi_{ij}(\mu_t) A_t / P$ for every i and j.
5. History: drop the oldest action in μ_t, append $-\text{sign}\, A_t$.
6. Price (extension): draw $Z_{t+1} \sim \mathcal{N}(0,1)$; $\ln S_{t+1} \leftarrow \ln S_t + A_t/\lambda + \sigma_S \sqrt{\Delta t}\, Z_{t+1}$.

Output: $A_0, \dots, A_{R-1}$, $S_0, \dots, S_R$, and $\sigma_A^2/N = \langle A^2 \rangle / N$.

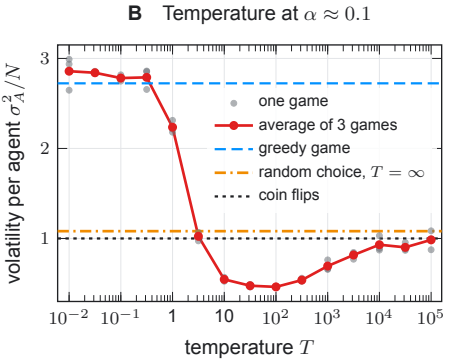
The Phase Diagram

Volatility per agent against α in the greedy game with $J = 2$. The game is crowded below $\alpha \approx 0.2$. The minimum is 0.14 at $\alpha \approx 0.35$, and above that the volatility rises toward 1 from below:



The Effect of Temperature

At $\alpha \approx 0.1$: greedy 2.72, random choice 1.08, and 0.46 at $T = 100$, below both limits. Our scores are divided by P, so our T is $1/\alpha$ times that of Cavagna and coauthors:



Optional Advanced Material

Maximum-entropy choice: advanced/max_entropy/
CHEME-5660-L12a-Derivation-MaxEntropyChoice-Fall-2026.ipynb

We prove the Boltzmann choice rule with Lagrange multipliers and relate the temperature to thermodynamics. A derivation with no code.

Biased policies: advanced/biased_policies/
CHEME-5660-L12a-Derivation-BiasedPolicies-Fall-2026.ipynb

What if the policies lean toward buying? We derive how much of the bias agents who follow their scores can remove, and show that above a threshold the market locks into a steady trend. A derivation with no code.

Summary

We introduced agent-based models, used the Boltzmann choice rule, the maximum-entropy choice, and connected trading rules to market-wide behavior.

Key takeaways

- Agent-based models: agents, information, a decision rule, an aggregate, and feedback.
- Maximum entropy: $p_j \propto \exp(U_j/T)$ has the largest entropy for a given expected score; T moves the agent from greedy to random.
- Micro to macro: Farmer–Joshi traders produce fat tails and clustering, but not every stylized fact at once; the minority game's volatility depends on α and T, and a crowded game moves the price more.